Practice Set 15 Paper 2 – Urology

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Practice paper compiled from the question bank.

Q1. Describe classification and management of Wilm's bladder

Type: LAQ (25 marks) (bilateral Wilms tumour also answered in 2020-Summer Paper 3, Q3e and 2016-Summer Paper 3, Q3c; Wilms tumour in Practice Set 10 Paper 2, Q3b)

Read as: "Wilms tumour" (nephroblastoma); "bladder" is taken as a typo.

Introduction

- WT (Wilms tumour) is an embryonal tumour of metanephric blastema. It forms 95% of kidney cancers under 15 years; median age 3.5 years; over 80% present before 5 years.
- **Genetics:** WT1 (11p13), IGF2/11p15 (commonest change, about 70%), WTX, CTNNB1; **TP53 in most anaplastic tumours**. Familial 1–2%.
- **Syndromes:** WAGR (Wilms, aniridia, genitourinary anomalies, intellectual disability) and Denys–Drash (46,XY DSD [disorder of sex development], mesangial sclerosis): risk 50%. Beckwith–Wiedemann and hemihypertrophy: 5–10%. Screen by ultrasound every 3–4 months until 7 years when risk is >1%.
- **Nephrogenic rests** (precursors) occur in over one-third of WT kidneys and over 90% of bilateral cases.

Classification

A. Histological

Group	Features	Significance
Favourable histology	Triphasic: blastema, epithelium (tubules), stroma; pseudocapsule	Survival >90%
Anaplasia (unfavourable)	Nuclei ≥3× neighbours, hyperchromasia, abnormal mitoses; **focal** or **diffuse** (worse)	Chemoresistant; poor even at stage I

- **SIOP (International Society of Paediatric Oncology) post-chemotherapy risk groups:**

Risk	Histology
Low	Completely necrotic
Intermediate	All others (regressive, mixed, epithelial, stromal, focal anaplasia)
High	**Blastemal-predominant**; diffuse anaplasia

- Residual blastema predicts relapse; stromal and epithelial tumours shrink little but are cured by excision.

B. Staging: COG (Children's Oncology Group), surgical-pathological, before chemotherapy

Stage	Description
I	Confined to kidney, completely resected, capsule intact, no renal sinus vessel invasion
II	Beyond kidney (renal sinus vessels, perirenal fat, vein thrombus) but completely resected
III	Residual abdominal disease: **positive nodes, any spill, prior biopsy**, positive margin, peritoneal implants
IV	Haematogenous metastases (lung commonest, liver, bone, brain)
V	Bilateral at diagnosis (each side staged separately)

- **SIOP staging** uses the same five stages but is applied **after preoperative chemotherapy**. Necrotic, chemotherapy-induced change in the sinus or perirenal fat does not upstage.

- **Biology:** combined **1p and 16q LOH (loss of heterozygosity)** predicts relapse; 1q gain is the next stratifier.

Evaluation

- Abdominal mass; haematuria about 20%; hypertension about 25% (renin). Varicocele or hepatomegaly suggests IVC (inferior vena cava) thrombus. Acquired von Willebrand disease in up to 8%.
- Ultrasound with Doppler → contrast CT (computed tomography) or MRI abdomen → **chest CT** (before anaesthesia). Blood count, renal and liver function, coagulation.

Management: two philosophies

	COG (North America)	SIOP (Europe)
Sequence	**Upfront radical nephrectomy**, then chemotherapy	**Preoperative VCR + AMD** (4 weeks; add DOX for 6 weeks if metastatic), no biopsy, then nephrectomy
Advantage	Untreated histology and biology for risk assignment	Tumour shrinks; spill almost zero
Risk basis	Stage + histology + LOH	Post-chemotherapy histology + stage

(VCR = vincristine; AMD = actinomycin D; DOX = doxorubicin.)

Surgical principles (radical nephrectomy)

- Transperitoneal approach; palpate renal vein and IVC before ligation; spare a tumour-free adrenal.
- **Avoid spill** (it makes the tumour stage III).
- **Lymph node sampling of at least 5 nodes** (hilar, paracaval, para-aortic, interaortocaval) is essential; formal dissection is not. Omitted sampling is the commonest protocol violation.
- IVC thrombus at or above the hepatic veins → preoperative chemotherapy to avoid bypass.

Postoperative therapy (COG, favourable histology)

Group	Treatment
Very low risk (<2 years, <550 g, stage I)	Nephrectomy alone (AREN0532)
Stage I–II	VCR + AMD
Stage III	VCR + AMD + DOX + flank RT (radiotherapy)
Stage IV	Three drugs + whole-lung RT; lung RT omitted if nodules clear by week 6 (AREN0533)
Anaplasia or 1p/16q LOH	Intensified regimens (regimen M adds cyclophosphamide + etoposide)

Bilateral WT (stage V) and nephron sparing

- Bilateral WT occurs in 5–7%. **No upfront nephrectomy or biopsy.** The aim is cure while preserving nephrons.

```
VCR + AMD + DOX × 6 weeks → reimaging
  ├─ resectable → bilateral NSS (nephron-sparing surgery)
  └─ not → 6 more weeks → surgery by week 12
       poor response → bilateral open biopsy
```

- **Why by week 12:** in NWTS-4 (National Wilms Tumor Study-4) prolonged chemotherapy delayed surgery and some tumours became anaplastic.
- **AREN0534:** 4-year EFS (event-free survival) 82.1% and OS (overall survival) 94.9%.
- Bilateral nephrectomy and dialysis are rare; transplant after 1–2 years.

- **Unilateral NSS:** for syndromic children and solitary kidney; SIOP UMBRELLA criteria are volume <300 mL, a substantial remnant and negative nodes.

Relapse and late effects

- Salvage survival: 70–80% after VCR/AMD only; about 10% for anaplastic or blastemal relapse.
- **Late effects:** cardiotoxicity (DOX), second cancers (RT), infertility, renal failure (bilateral).

Diagram to draw: COG staging diagram showing capsule, renal sinus, nodes and spill.

Key points to remember:

- Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
- Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
- COG operates first; SIOP gives VCR + AMD first.
- Bilateral: three drugs, surgery by week 12, NSS.

Q2. Discuss management of organ confined CA prostate

Type: LAQ (25 marks)

Introduction

- Organ-confined prostate cancer is **cT1–T2 N0 M0**. Treatment must balance cancer mortality against **competing mortality**: life expectancy, comorbidity, baseline urinary, sexual and bowel function, and patient preference (shared decision-making).

Risk stratification (NCCN, National Comprehensive Cancer Network)

Risk group	Criteria
Very low	All of: T1c, GG (grade group) 1, PSA (prostate-specific antigen) <10 ng/mL, <3 positive cores with ≤50% cancer in each, PSA density <0.15
Low	All of: T1–T2a, GG1, PSA <10
Favourable intermediate	1 intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive
Unfavourable intermediate	2–3 intermediate factors, GG3, or ≥50% cores positive
High	Any of: T3a, GG4–5, PSA >20

- **Grade groups:** GG1 = 3+3; GG2 = 3+4; GG3 = 4+3; GG4 = 8; GG5 = 9–10.

Evaluation

- PSA, DRE (digital rectal examination), mpMRI (multiparametric MRI) with targeted and systematic biopsy.
- **Staging imaging** from unfavourable intermediate risk: bone scan + CT/MRI, or PSMA (prostate-specific membrane antigen) PET (positron emission tomography). **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Life expectancy (life tables, Charlson index); IPSS (International Prostate Symptom Score) and erectile function.
- **Also know:** Partin tables, Kattan nomogram, CAPRA (Cancer of the Prostate Risk Assessment) score; genomic tests (Decipher, Prolaris, Oncotype DX) for borderline surveillance decisions.

Options by risk

Risk	Preferred options
Very low / low	**Active surveillance (AS)**; watchful waiting if life expectancy short
Favourable intermediate	AS in selected men; RP (radical prostatectomy); EBRT (external-beam radiotherapy) or brachytherapy alone
Unfavourable intermediate	RP + PLND (pelvic lymph node dissection); EBRT + **4–6 months ADT** (androgen deprivation therapy)
High (organ-confined, e.g. GG4–5 or PSA >20)	RP + extended PLND (multimodal); EBRT + **18–36 months ADT**

1. Active surveillance vs watchful waiting

- **AS:** deferred treatment with **curative intent**. Monitor by PSA every 3–6 months, DRE, **confirmatory biopsy or MRI within 6–12 months**, then periodic MRI and biopsy. Trigger for treatment: grade progression (GG ≥2–3), increasing volume or patient choice.

- Entry criteria: Johns Hopkins (strict, Epstein), PRIAS (Prostate Cancer Research International Active Surveillance), Toronto (inclusive). About half are treated by 10–15 years.
- **Watchful waiting:** no curative intent; ADT for symptomatic progression. For limited life expectancy.

2. Radical prostatectomy

- Open retropubic or **robot-assisted** (similar outcomes in the randomised trial). Nerve sparing when there is no same-side extracapsular risk.
- **PLND** when nodal risk is significant on nomograms (extended template).
- **Complications:** incontinence, ED (erectile dysfunction), anastomotic stricture, rectal injury.
- **Advantages:** pathological staging; relieves obstruction; RT kept for salvage.

3. Radiotherapy

- **EBRT:** IMRT (intensity-modulated RT) with image guidance. Fractionation: conventional 78 Gy in 39; **moderate hypofractionation 60 Gy in 20** (CHHiP: non-inferior); **SBRT (stereotactic body RT) 36.25 Gy in 5** (PACE-B: similar toxicity).
- **Why add ADT:** synergistic cell kill and target shrinkage. **RTOG 94-08:** 4 months ADT improved OS in intermediate risk; **DFCI 95-096:** 6 months ADT improved 8-year OS (61% → 74%).
- **Brachytherapy:** LDR (low-dose-rate) seeds for low and favourable intermediate risk. Relative contraindications: gland >60 mL, high IPSS, recent TURP. HDR (high-dose-rate) or LDR **boost** with EBRT for higher risk (ASCENDE-RT: half the biochemical failure).
- **Toxicity:** proctitis, irritative urinary symptoms, late ED, second cancers (controversial).

4. Focal and whole-gland ablation

- HIFU (high-intensity focused ultrasound), cryotherapy, focal laser, IRE (irreversible electroporation), VTP (vascular-targeted photodynamic therapy).
- Based on the **index lesion hypothesis**. Best for MRI-visible, biopsy-confirmed, unifocal **intermediate-risk (GG2)** disease. Investigational; ideally in trials. VTP is the only modality with a randomised trial (versus AS in low risk).

Landmark trials

Trial	Comparison	Result
SPCG-4 (pre-PSA era)	RP vs watchful waiting	Cancer death RR (relative risk) 0.56 at 23 years; NNT 8; benefit <65 years
PIVOT	RP vs observation (PSA era)	No mortality difference; benefit only in intermediate risk
ProtecT	Monitoring vs RP vs RT + ADT	15-year cancer mortality 2.7%, 2.2%, 2.9% (no difference); more metastases with monitoring

- **ProtecT function:** RP worst for continence and sexual function; RT transient bowel effects.

Algorithm

```
Organ-confined CaP → risk group + life expectancy
 ├─ short life expectancy → watchful waiting
 ├─ very low / low → active surveillance
 ├─ favourable intermediate → AS (selected), RP, EBRT, brachytherapy
 └─ unfavourable intermediate / high → RP + PLND
                                    or EBRT + ADT (4–6 or 18–36 months)
```

Follow-up

- PSA every 6–12 months. **Biochemical recurrence:** after RP, PSA ≥0.2 ng/mL confirmed; after RT, **nadir + 2 ng/mL (Phoenix)**.

Recent advances

- PSMA PET staging; MRI-led surveillance; SBRT; genomic classifiers.

Diagram to draw: stepwise decision tree (treat or not → partial or whole gland → surgery or radiation).

> **Key points to remember:**
> - Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
> - AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
> - EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
> - Focal therapy rests on the index lesion hypothesis; still investigational.

Q3a. Monopolar v/s Bipolar TURP.

Type: SAQ (10 marks)

Introduction

- TURP (transurethral resection of the prostate) removes transition-zone tissue from bladder neck to verumontanum, down to the surgical capsule; the benchmark BPH (benign prostatic hyperplasia) operation.
- **AUA (American Urological Association):** no preference; choose by expertise.

Principles

- **Monopolar (m-TURP):** current runs from the loop through the patient to a grounding pad. Saline would disperse the current, so a **non-conducting, hypo-osmolar irrigant** (glycine 1.5%, sorbitol–mannitol, water) is needed.
- **Bipolar (b-TURP):** active and return poles are on the same instrument, so current stays local and **normal saline** can be used. The Plasma system turns saline into a plasma layer (orange glow) that cuts at lower temperature and seals vessels.

Comparison

Feature	Monopolar	Bipolar
Current path	Loop → patient → pad	Within the loop
Irrigant	Glycine, sorbitol, water	Normal saline
TUR syndrome	Yes	Practically eliminated
Resection time	Limited (speed over haemostasis)	No time pressure; large glands
Loop	Tungsten	Platinum–iridium (Plasma)
Bleeding, transfusion, clot retention	Higher	Lower
Catheter time, hospital stay	Longer	Shorter
Obturator reflex	Risk	Reduced
IPSS (International Prostate Symptom Score), QoL, PVR	Equivalent (Qmax marginally better with bipolar)	Equivalent
Stricture, BNC (bladder neck contracture), sexual function, retreatment	Similar	Similar

TUR syndrome (monopolar)

- Irrigant absorbed through open sinuses (venous pressure about 10 mmHg) → **dilutional hyponatraemia**: confusion, visual disturbance, nausea, bradycardia, hypertension, seizures; glycine also yields ammonia.
- **Risk factors:** gland >45 mL, resection >90 minutes, high irrigation bag.
- **Treatment:** stop surgery, check sodium; 3% saline 100 mL boluses for neurological symptoms; avoid over-rapid correction (central pontine myelinolysis).

Evidence

- Meta-analyses: equal efficacy up to 5 years; b-TURP safer (NNT about 50 to prevent one TUR syndrome, about 20 to prevent one clot retention).
- **Also know:** TURis (sheath as return) caused strictures; bipolar enucleation and vaporisation (PlasmaButton).

Diagram to draw: monopolar circuit (loop → patient → pad) versus bipolar (loop to return electrode on the same resectoscope).

> **Key points to remember:**
>
> - Bipolar allows saline; TUR syndrome practically eliminated.
> - Equal efficacy; bipolar has less bleeding and shorter catheter time.
> - TUR syndrome risk: >45 mL gland, >90 minutes.

Q3b. Overactive bladder

Type: SAQ (10 marks) (OAB also answered in Practice Set 4 Paper 1, Q2; detrusor relaxants in Practice Set 1 Paper 1, Q3e)

Definition

- **OAB (overactive bladder), ICS (International Continence Society): urgency**, usually with frequency and nocturia, with or without UUI (urgency urinary incontinence), in the absence of UTI (urinary tract infection) or other obvious pathology.
- **OAB dry** vs **wet** (with UUI). OAB is a symptom diagnosis; DO (detrusor overactivity) is urodynamic, found in only 44–69% with urgency.

Pathophysiology

- **Afferent:** silent C fibres become active and hypersensitive.
- **Urothelial:** urothelium releases ATP (adenosine triphosphate) and acetylcholine onto suburothelial afferents.
- **Myogenic:** patchy denervation → spontaneous micromotions that spread.
- **Central:** loss of forebrain inhibition of the pontine micturition centre.
- **Associations:** ageing, BOO (bladder outlet obstruction), metabolic syndrome.

Evaluation

- **Minimum:** history, examination, **urinalysis**. Bladder diary (3 days), questionnaires, PVR (post-void residual) when indicated.
- Urodynamics and cystoscopy only for complicated or refractory cases.

Management (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction]; not a compulsory sequence)

Line	Treatment
Behavioural	Education; reduce fluids by about 25%; cut caffeine; weight loss; **bladder training** (timed voids, interval increased by 15–30 minutes); **PFMT (pelvic floor muscle training)** with urge suppression: 50–80% fewer leaks
Drugs	**Antimuscarinics** (oxybutynin, solifenacin, extended-release trospium, darifenacin, fesoterodine); **β3-agonists** (mirabegron, vibegron); combinations
Third line	**PTNS (percutaneous tibial nerve stimulation)** weekly × 12; **onabotulinumtoxinA 100 U** intradetrusor (about 25% dry; UTI about 20%; retention); **SNM (sacral neuromodulation)** at S3 (success about 70–80%; revision >30% at 5 years)
Last resort	Augmentation cystoplasty or urinary diversion

- **Antimuscarinic use >3 months raises dementia risk** → prefer β3-agonists in the elderly. Men: α-blocker first; add antimuscarinic if PVR is low.
- **Trials:** ROSETTA (Botox beat SNM at 6 months, more UTI); SUmiT (PTNS 55% vs sham 21%).

Diagram to draw: bladder showing urothelial afferents and C fibres with brainstem and cortical inhibition.

Key points to remember:

- OAB is defined by urgency; DO is urodynamic.
- Behavioural therapy first; antimuscarinic or β3-agonist next.
- Botox 100 U, PTNS, SNM for refractory OAB.
- Long-term antimuscarinics → dementia risk.

Q3c. Fracture penis

Type: SAQ (10 marks)

Definition and mechanism

- **Penile fracture** is a tear of the **tunica albuginea** with rupture of the corpus cavernosum, in the **erect** penis.
- Abnormal bending of the erect penis causes an abrupt rise in intracavernosal pressure beyond tunical strength → **transverse tear**.
- The tunica (inner circular, outer longitudinal layers) is **thinnest ventrolaterally**, so most tears are ventral or lateral, proximal to mid shaft.
- **Causes:** intercourse (commonest; rear-entry), masturbation, rolling over; self-inflicted "taqaandan" (Middle East); collagenase injection for Peyronie disease.
- **Bilateral tears in about 10%**; urethral injury about 20% in the West vs about 3% in Asia and the Middle East.

Clinical features

- **Cracking or popping sound → pain → rapid detumescence → swelling and discoloration.**
- **"Eggplant deformity"** when Buck's fascia is intact; if torn, haematoma spreads to scrotum, perineum and suprapubic area.
- Penis deviates **away from the tear**; tunical defect may be palpable.
- **Urethral injury:** blood at meatus, haematuria, retention.

Investigations

- **Diagnosis is clinical** (AUA [American Urological Association] Urotrauma Guideline).
- **Ultrasound** for equivocal cases (localises the tear); **MRI** (magnetic resonance imaging) if still equivocal (100% negative predictive value). Cavernosography is discouraged.
- Urethra: **intraoperative flexible cystoscopy** (AUA and EAU [European Association of Urology] preference) or RUG (retrograde urethrogram).
- **"False fracture":** dorsal vein or artery rupture.

Management

- **Immediate surgical repair** (less curvature, ED [erectile dysfunction] and nodules than conservative care).

1. Incision: ventral penoscrotal or lateral over the defect; subcoronal degloving if site unclear (preputial ischaemia risk).
2. Evacuate haematoma; find the tear (artificial erection helps).
3. Close the tunica with **interrupted 2-0/3-0 absorbable sutures**; avoid debriding erectile tissue.
4. Urethra: partial → close over catheter; complete → tension-free repair; catheter 2–3 weeks.

- Abstinence about 1 month. Delay up to 7 days does not worsen outcome. After collagenase injection without urethral injury, conservative care is acceptable.

Complications

- About 20%: nodule, curvature, ED, painful erection, infection, stricture or fistula.

Diagram to draw: cross-section of penis showing tunical tear, Buck's fascia and haematoma spread.

> **Key points to remember:**
>
> - Snap, detumescence, eggplant deformity.
> - Urethral injury about 20% (West); suspect if bilateral.
> - Immediate exploration and interrupted absorbable tunical closure.

Q3d. Describe various penile prosthesis in ED

Type: SAQ (10 marks) (also answered in 2023-Winter Paper 4, Q3c and 2018-Summer Paper 2, Q3d; surgery for ED in 2020-Summer Paper 2, Q2)

Introduction

- Most effective, but last-line, treatment of ED (erectile dysfunction); **satisfaction 80–90%**. First devices: Scott inflatable (1972), Small–Carrion rod (1978).
- **Indications:** failure of, contraindication to or refusal of PDE5Is (phosphodiesterase type 5 inhibitors), injections and vacuum devices; ED with Peyronie disease; after ischaemic priapism (early implant preserves length).
- **Contraindications:** active infection, uncontrolled diabetes, unrealistic expectations.

Types

Type	Devices	Features
Malleable (semirigid) rods	Tactra (nitinol core), Genesis (braided silver), Rigi 10 (stainless steel)	Two bendable rods; simple, cheap; mechanical failure exceedingly rare; permanently firm; suited to poor dexterity
Mechanical rods	Spectra (discontinued)	Articulating segments
Two-piece inflatable	Ambicor (discontinued)	Cylinders + scrotal pump; no reservoir; poor flaccidity
Three-piece inflatable	AMS 700 CX/LGX/CXR; Coloplast Titan, Titan Touch, Titan Narrow; Rigicon Infla 10	Two cylinders, scrotal pump, abdominal reservoir; **best rigidity and true flaccidity**; most popular

Three-piece inflatable details

- **Expansion:** girth only (CX, Titan) or girth and length (LGX, up to 25%; Infla 10 AX). **Narrow** cylinders (CXR, Titan Narrow) for fibrotic corpora.
- **Material:** AMS and Rigicon use silicone–fabric–silicone layers; Titan uses polyurethane (Bioflex).
- **Coatings:** InhibiZone (rifampicin + minocycline) on AMS; hydrophilic coating on Titan, dipped in antibiotic.
- **Reservoir:** space of Retzius, or **ectopic submuscular** after pelvic surgery; lockout valve prevents autoinflation.

Surgical principles

- **Approach:** penoscrotal (commonest) or infrapubic (dorsal nerve at risk).
- Corporotomy → dilate (instruments dorsolateral) → check for crossover or perforation → measure → cylinders sized to corporal length. Rods are placed slightly shorter.
- Antibiotics, glucose control, no-touch technique.

Complications

- **Infection 1–3%** (biofilm): remove all components; **Mulcahy salvage** (washout and immediate replacement).
- **Mechanical failure** (inflatables): repair about 15% at 5 years, 30% at 10.
- Crural perforation, urethral injury, erosion, glans necrosis, shortening, autoinflation.

Diagram to draw: three-piece inflatable: cylinders, scrotal pump, reservoir in the space of Retzius.

> **Key points to remember:**
> - Three-piece inflatable: best rigidity and flaccidity.
> - Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
> - Infection 1–3%; satisfaction 80–90%.

Q3e. Ovarian vein syndromes

Type: SAQ (10 marks)

Definition

- **Ovarian vein syndrome** (Clark, 1964) is extrinsic compression of the **right ureter** by a dilated or thrombosed **right ovarian vein**, causing intermittent hydroureteronephrosis. It is a **controversial** entity; many cases are now attributed to other causes.
- The plural also covers ovarian vein thrombosis and pelvic congestion.

Anatomy (basis)

- Ovarian veins run in the infundibulopelvic ligaments. The **right** enters the IVC (inferior vena cava) obliquely; the **left** enters the left renal vein at a right angle.
- Gonadal vessels **cross anterior to the ureter**; ovarian veins dilate greatly in pregnancy.

Types

Syndrome	Mechanism	Features
Ovarian vein (ureteric) syndrome	Dilated right ovarian vein compresses the ureter at the crossing	Multiparous women; right flank pain (premenstrual or in pregnancy); recurrent UTI (urinary tract infection)
Ovarian vein thrombosis	Postpartum hypercoagulability and venous stasis; mostly right-sided	Fever and right lower-quadrant or flank pain after delivery; may obstruct ureter; risk of pulmonary embolism
Pelvic congestion syndrome	Incompetent, refluxing ovarian veins with pelvic varices; sometimes secondary to nutcracker (left renal vein compression)	Chronic pelvic pain worse on standing, dyspareunia, post-coital ache

Controversy

- Right-sided dilatation in pregnancy is mostly **physiological hydronephrosis** (progesterone plus uterine compression; stops at the pelvic brim). Venous compression is rarely proven, so diagnosis is by **exclusion**.

Investigations

- Ultrasound; **CT urography** or IVU (intravenous urography): right hydroureter to a smooth oblique extrinsic indentation, normal ureter below.
- **Diuretic renogram** to prove obstruction; retrograde pyelography.
- CT or MR (magnetic resonance) venography, Doppler; **gonadal venography** (gold standard for reflux).
- **Exclude:** stone, retrocaval ureter, iliac vessel crossing, retroperitoneal fibrosis, endometriosis.

Management

- **Asymptomatic:** observe. **Obstruction:** stent or nephrostomy if infected or pregnant; definitive **ureterolysis with ligation and division of the ovarian vein** (open or laparoscopic).
- **Thrombosis:** anticoagulation and antibiotics.
- **Pelvic congestion:** progestogens or GnRH (gonadotropin-releasing hormone) analogues; **coil or sclerotherapy embolisation**; ligation.

Diagram to draw: right ovarian vein crossing anterior to the right ureter, with proximal hydroureteronephrosis.

Key points to remember:

- Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
- Postpartum ovarian vein thrombosis: right side, anticoagulate.
- Treatment: ureterolysis with vein division, or embolisation.

Q3f. Megameatus intact prepuce

Type: SAQ (10 marks)

Definition

- **MIP (megameatus intact prepuce)** is a hypospadias variant (up to 5%) with a **wide, "fish-mouth" glanular or coronal meatus**, a **completely formed prepuce** and **no significant curvature**. Described by Duckett and Keating (1989).

Embryology

- Normally the prepuce closes ventrally only after the urethral folds fuse (about 20 weeks), so 95% of hypospadias have a hooded, incomplete foreskin.
- In MIP the foreskin is complete, showing that **preputial and urethral development can dissociate**.

Clinical features

- Hidden under a normal foreskin; often found on **retraction before circumcision** or during circumcision.
- Wide, deep glans groove with a large, patulous meatus; well-developed spongiosum.
- Symptoms: spraying or deflected stream; cosmetic concern.
- **Advice:** retract the foreskin before every newborn circumcision. If found, the circumcision may be completed; prior circumcision does not harm later repair.

Evaluation

- Examine meatal site and size, glans and plate, curvature (artificial erection at surgery). GMS (glans–meatus–shaft) score.
- No routine upper-tract imaging for isolated distal hypospadias.

Management

- **Indication:** cosmesis and a straight stream; repair is optional. Usual timing 6–18 months.
- **Pyramid procedure (Duckett and Keating):**
  1. Glans traction suture; retract prepuce.
  2. Incise around the megameatus in a **pyramid shape**, apex on the ventral shaft.
  3. Dissect the wide distal urethra free of the glans wings.
  4. **Excise a ventral wedge** of redundant urethra to taper it.
  5. Tubularise over a catheter; cover with dartos.
  6. Approximate glans wings in two layers (glansplasty); preserve or reconstruct the prepuce.
- **Alternatives:** GAP (glans approximation procedure; pyramid = GAP with intact prepuce), TIP (tubularised incised plate) urethroplasty, Thiersch–Duplay tubularisation, Mathieu.

Complications

- Fistula, meatal stenosis or regression, glans dehiscence; generally low after distal repair.

Diagram to draw: fish-mouth glanular meatus with intact prepuce, and the pyramid incision.

> **Key points to remember:**
> - Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
> - Often found at circumcision; retract foreskin first.
> - Pyramid repair: dissect, taper and tubularise; glansplasty.

Rapid Revision Sheet – Practice Set 15 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe classification and management of Wilm's bladder (LAQ)

- Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
- Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
- COG operates first; SIOP gives VCR + AMD first.
- Bilateral: three drugs, surgery by week 12, NSS.

Q2. Discuss management of organ confined CA prostate (LAQ)

- Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
- AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
- EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
- Focal therapy rests on the index lesion hypothesis; still investigational.

Q3a. Monopolar v/s Bipolar TURP. (SAQ)

- Bipolar allows saline; TUR syndrome practically eliminated.
- Equal efficacy; bipolar has less bleeding and shorter catheter time.
- TUR syndrome risk: >45 mL gland, >90 minutes.

Q3b. Overactive bladder (SAQ)

- OAB is defined by urgency; DO is urodynamic.
- Behavioural therapy first; antimuscarinic or β3-agonist next.
- Botox 100 U, PTNS, SNM for refractory OAB.
- Long-term antimuscarinics → dementia risk.

Q3c. Fracture penis (SAQ)

- Snap, detumescence, eggplant deformity.
- Urethral injury about 20% (West); suspect if bilateral.
- Immediate exploration and interrupted absorbable tunical closure.

Q3d. Describe various penile prosthesis in ED (SAQ)

- Three-piece inflatable: best rigidity and flaccidity.
- Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
- Infection 1–3%; satisfaction 80–90%.

Q3e. Ovarian vein syndromes (SAQ)

- Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
- Postpartum ovarian vein thrombosis: right side, anticoagulate.
- Treatment: ureterolysis with vein division, or embolisation.

Q3f. Megameatus intact prepuce (SAQ)

- Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
- Often found at circumcision; retract foreskin first.
- Pyramid repair: dissect, taper and tubularise; glansplasty.